

Syamchand Chilaka
Data Scientist | AI/ML Engineer

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Professional Summary

Senior AI/ML Engineer and Data Scientist with 5+ years of experience building and scaling production GenAI and machine learning systems in financial services and consulting. Proven track record deploying RAG architectures, distributed training/inference infrastructure, and enterprise AI tools across highly regulated environments. Specialized in managing engineering trade-offs between computational throughput, inference latency, and enterprise compliance constraints.

Skills

- **GenAI & Inference:** LLM apps (LLaMA, BERT), RAG, multi-agent workflows, LangGraph/LlamaIndex, vLLM, TensorRT-LLM, vector DBs (Pinecone, Qdrant, pgvector), tool/function calling.
- **ML & Vision:** PyTorch, Hugging Face, XGBoost, CNNs, anomaly detection, recommender systems, OpenCV, A/B testing.
- **Data & Compute:** Databricks, Apache Spark (PySpark), Ray, feature stores, SQL, AWS (EKS, EC2, S3), Docker, Kubernetes.
- **MLOps & Monitoring:** MLflow, Airflow, Jenkins CI/CD, Prometheus, Grafana, LlamaGuard.
- **Languages:** Python, Java, Bash.

Work Experience

Citi - Data Scientist / AI Engineer | New Jersey | Jan 2024 – Present

- **Infrastructure Optimization:** Led the design and deployment of a vLLM- and TensorRT-LLM-based inference stack for internal LLM workloads, optimizing KV-cache utilization via semantic prompt caching to reduce p95 latency by 180 ms and monthly token spend by 40%.
- **Agentic System Design:** Partnered with internal operations and product teams to build a stateful multi-agent routing workflow using OpenAI and Anthropic function calling, resolving manual triage bottlenecks and reducing operational triage volume by 35%.
- **Enterprise RAG Scale:** Designed and implemented a production RAG pipeline over a 10M+ multi-format financial document corpus using semantic chunking and Cross-Encoder reranking, improving automated retrieval precision by 26%.
- **Distributed Fine-Tuning:** Deployed distributed fine-tuning pipelines (PEFT/LoRA) for open-weights models (LLaMA, Mistral) using Ray on AWS compute clusters, adapting models to domain-specific terminology under strict financial compliance constraints.
- **LLMOps & Observability:** Built and deployed centralized LLM observability pipelines using Prometheus, Grafana, and LlamaGuard to actively monitor latency, data drift, and safety signals across live production workflows.
- **Technical Leadership:** Led a sub-team of engineers to build and deploy stateful LangGraph multi-agent routing workflows, partnering with infrastructure and product teams to safely scale high-concurrency internal financial operations.

Accenture - Data Scientist II / Machine Learning Engineer | India | Sep 2020 – Jul 2022

- **Edge Vision Frameworks:** Designed and deployed edge-based CNN anomaly and defect detection frameworks for industrial client systems, achieving a 95% classification accuracy on active production lines.
- **Distributed Systems Compute:** Built distributed data processing and inference pipelines on Databricks and Apache Spark, accelerating data preprocessing execution speeds by 30% for multi-terabyte enterprise datasets.
- **Core Predictive Modeling:** Developed predictive machine learning systems and deep ranking models utilizing gradient boosting (XGBoost) and vector embeddings, driving a 22% lift in user engagement and a 20% increase in click-through rates.
- **Cross-Functional Delivery:** Partnered with corporate stakeholders and product leads to translate ambiguous business bottlenecks into production-ready ML microservices, shortening evaluation-to-deployment cycles by 25%.
- **Engineering Mentorship:** Led an engineering squad to design and productionize distributed training pipelines on Databricks, establishing standardized code reviews and mentoring junior developers to cut client model-deployment cycles by 25%.
- **Production MLOps:** Implemented containerized model deployments using Docker and Kubernetes (EKS), establishing automated Jenkins CI/CD pipelines for post-deployment performance tracking and drift monitoring.

Accenture - Junior Data Scientist | India | Jan 2020 – Aug 2020

- **Predictive Analytics:** Engineered predictive modeling pipelines (Decision Trees, Random Forests, XGBoost) with advanced feature engineering, achieving 72% accuracy in customer response classification.
- **Computer Vision Deployments:** Built a computer vision defect detection model (CNNs) deployed at the edge with 95% accuracy.
- **Statistical Optimization:** Designed and executed A/B testing frameworks in Python to optimize conversion metrics by 20%, while automating SQL and Python ETL pipelines to cut data preparation time by 20%.
- **CI/CD Automation:** Automated CI/CD pipelines with Git and Jenkins on Azure container environments, reducing deployment cycle times by 15%.

Projects

- **Project Axis - AI Red Teaming & Evaluation | Handshake AI x Stanford HAI:** Task Author on a Stanford HAI research initiative evaluating frontier AI models; designed PhD-level reasoning questions grounded in recent arXiv papers to expose traceable failure modes, built gold-standard solutions with step-by-step calculations, and diagnosed reasoning errors across multiple frontier models through an automated evaluation pipeline.
- **Project Helix - Software Engineering Fellow | Handshake AI:** Contributed to the training and evaluation of LLM coding assistants by authoring high-quality code samples, reviewing model-generated solutions across Python, SQL, and ML workflows, and providing structured feedback to improve code generation, reasoning, and debugging behavior.

Education

Master of Science in Computer and Information Science | **Long Island University, Brooklyn, New York**
Bachelor of Technology in Computer Science | **REVA University, Bangalore, Karnataka, India**

Certifications

- Certified Artificial Intelligence Governance Professional (IAPP AIGP)
- Model Validation 1 (Trainer)
- Model Validation 2 (Expert)
- Microsoft Responsible AI Certification